out; and instead of beholding abrupt mountain-axes of rock solidified under great pressure, deluges of lava would have flowed out at innumerable points on every line of elevation. [2]

[1] M. Arago in L'Institut, 1839, p. 337. See also Miers's Chile, vol. i. p. 392; also Lyell's Principles of Geology, chap. xv., book ii.

[2] For a full account of the volcanic phenomena which accompanied the earthquake of the 20th, and for the conclusions deducible from them, I must refer to Volume V. of the Geological Transactions.

CHAPTER XV

PASSAGE OF THE CORDILLERA

Valparaiso—Portillo Pass—Sagacity of Mules—Mountain-torrents—Mines, how discovered—Proofs of the gradual Elevation of the Cordillera—Effect of Snow on Rocks—Geological Structure of the two main Ranges, their distinct Origin and Upheaval—Great Subsidence—Red Snow—Winds—Pinnacles of Snow—Dry and clear Atmosphere—Electricity—Pampas—Zoology of the opposite Side of the Andes—Locusts—Great Bugs—Mendoza—Uspallata Pass—Silicified Trees buried as they grew—Incas Bridge—Badness of the Passes exaggerated—Cumbre—Casuchas—Valparaiso.

MARCH 7th, 1835.—We stayed three days at Concepcion, and then sailed for Valparaiso. The wind being northerly, we only reached the mouth of the harbour of Concepcion before it was dark. Being very near the land, and a fog coming on, the anchor was dropped. Presently a large American whaler appeared alongside of us; and we heard the Yankee swearing at his men to keep quiet, whilst he listened for the breakers. Captain Fitz Roy hailed him, in a loud clear voice, to anchor where he then was. The poor man must have thought the voice came from the shore: such a Babel of cries issued at once from the ship—every one hallooing out, "Let go the anchor! veer cable! shorten sail!" It was the most laughable thing I ever heard. If the ship's crew